



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to TPC FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

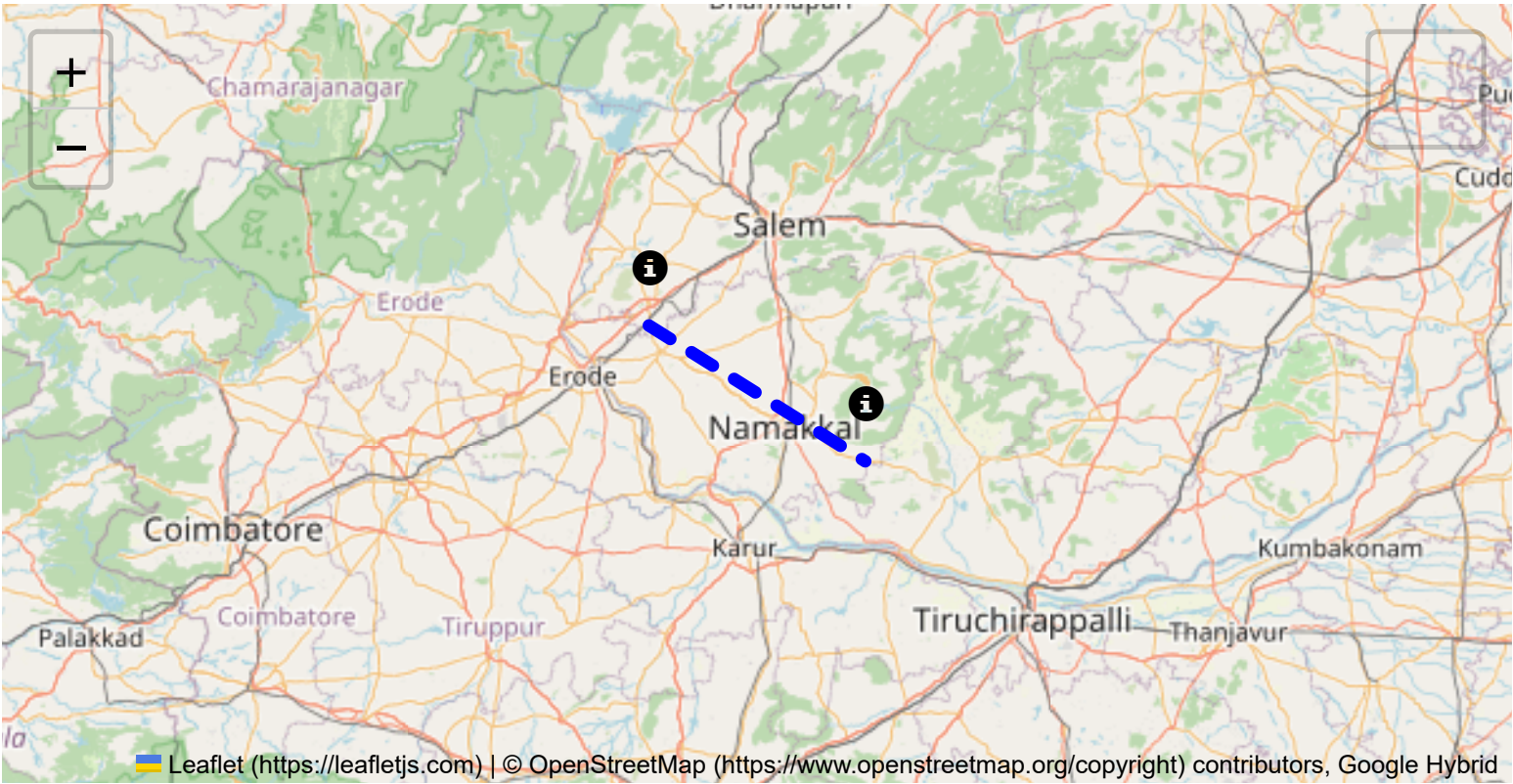
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

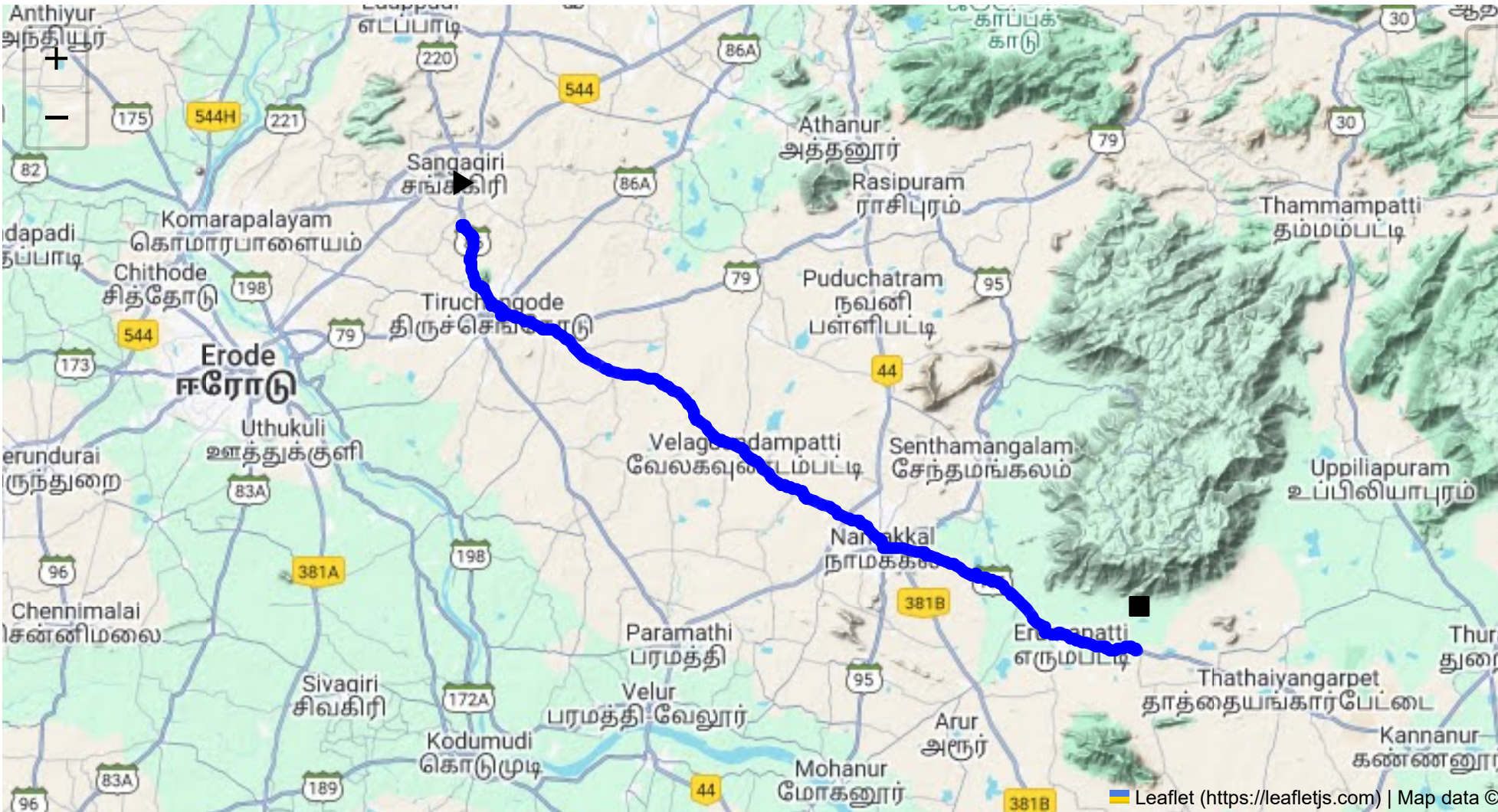
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 66.43 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (11.148334, 78.344806)



Welcome to the Journey Risk Management Study

To provide a comprehensive analysis of the route from CVQF+23W, Sangagiri to 48XW+5C7, Varagur in Tamil Nadu, India, here's a detailed assessment covering the requested points:

1. **Overview of the Route Map:** The journey from Sangagiri to Varagur traverses through rural and semi-urban areas in Tamil Nadu. The primary roads involved include National and State Highways with

some local roads near villages. The route offers a mix of straight highways and winding local roads, interspersed with small towns and agricultural fields.

2. **Typical Weather Conditions and Potential Weather-related Hazards:** Tamil Nadu experiences a tropical climate with hot and dry conditions. The monsoon season (June to October) may bring heavy rains, causing potential flooding or waterlogging on roads. The dry season can cause dust storms, reducing visibility.
3. **Analysis of Traffic Patterns:** The major roads may experience high traffic during peak hours, especially in the mornings (7-9 AM) and evenings (5-7 PM). Congestion-prone areas include entry and exit points of larger towns. Agricultural transport and local market days may increase traffic unpredictably.
4. **Assessment of Road Quality and Infrastructure:** National and State Highways generally offer good pavement quality, but local roads may have potholes and uneven surfaces. Bridges and culverts are mostly well-constructed, but caution is advised in identifying weight restrictions.
5. **Suggestions for Alternative Routes for Emergencies:** In case of emergencies, alternatives include taking parallel highways if available, or utilizing district and taluk roads to detour around congested or blocked sections. Detailed navigation apps can suggest live alternative routes.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Tamil Nadu mandates clear signage for vehicles carrying hazardous materials. Daytime driving is recommended to avoid nighttime visibility issues. Permits are required for specific materials; compliance with safety norms is imperative.
7. **Overview of Historical Incidents:** The area has seen incidents primarily related to overloading and vehicle failures. Accidents involving heavy vehicles are often due to mechanical failures or poor road conditions during adverse weather. Hazardous material incidents are less common but can arise from similar issues.
8. **Environmental Considerations and Sensitive Areas:** The route passes near agricultural zones, requiring careful attention to prevent spills or contamination. Certain sections may be close to protected areas; adherence to environmental regulations is critical to avoid harming local flora and fauna.
9. **Analysis of Communication Coverage:** Generally, major highways have good mobile coverage. However, remote stretches and sections closer to forests or farmlands may have signal interruptions. Pre-planning is advised to ensure communication capabilities throughout the journey.
10. **Estimated Emergency Response Times:** In urban and semi-urban areas, response times may be around 30-45 minutes. In rural stretches, it could be longer, around 60-90 minutes, depending on accessibility and emergency service availability.
11. **Overall Summary of Risk Assessment:** The route from Sangagiri to Varagur poses moderate risk due to potential weather-related road conditions, variability in traffic congestion, and inconsistent road quality. Caution is advised during monsoon season, and taking alternative routes when roads are congested is recommended. Managing hazardous materials requires adhering strictly to local regulations and being prepared for communication blackouts. The infrastructure generally supports

heavy vehicle traffic, but pre-trip inspections and continuous monitoring during the journey are crucial to ensuring safety.

Overall, the journey demands attention to environmental regulations and readiness for potential weather impacts, ensuring a diligent approach to safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	Medium	11.38342, 77.89975	30 KM/Hr
5	Turn	Medium	11.38295, 77.90001	30 KM/Hr
6	Turn	High	11.38186, 77.90258	15 KM/Hr
7	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
8	Turn	Medium	11.22407, 78.16640	30 KM/Hr
9	Turn	High	11.21816, 78.16934	15 KM/Hr
10	Turn	Medium	11.20105, 78.23198	30 KM/Hr
11	Turn	High	11.16124, 78.28095	15 KM/Hr
12	Turn	Medium	11.15975, 78.28885	30 KM/Hr
13	Turn	Medium	11.16047, 78.28943	30 KM/Hr
14	Turn	Medium	11.16056, 78.28971	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
16	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
19	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
20	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
21	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
22	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
23	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
24	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
25	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
26	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
27	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
28	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
29	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
31	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
32	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
33	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
34	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
35	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
38	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
42	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
44	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
45	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
46	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
47	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
48	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
49	hospital	SMK Hospital	11.2172759, 78.1706866	30 km/h	Medium
50	hospital	Manoharan Hospital	11.218016, 78.171629	30 km/h	Medium
51	hospital	Saravana Hospital	11.216652, 78.171298	30 km/h	Medium
52	hospital	Latha Eye Hospital	11.2185689, 78.1719035	30 km/h	Medium
53	hospital	Vasan Eye Care Hospital, Namakkal	11.216301, 78.1717694	30 km/h	Medium
54	hospital	Aishwarya Hospital	11.2164042, 78.1715972	30 km/h	Medium
55	hospital	Poornima Hospital	11.215533, 78.16911	30 km/h	Medium
56	hospital	Nagarajan Hospital	11.2171673, 78.1718618	30 km/h	Medium
57	clinic	Ramesh ENT & Skin Care	11.217133, 78.171071	30 km/h	Medium
58	clinic	ENT Clinic	11.215786, 78.170545	30 km/h	Medium
59	clinic	Hari Chest Clinic	11.2162053, 78.1710093	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
60	hospital	Dr. Kandasamy Hospital	11.216051, 78.172052	30 km/h	Medium
61	clinic	Dr. Shanmugam ENT Clinic	11.218762, 78.171206	30 km/h	Medium
62	clinic	Children's Clinic	11.217004, 78.171504	30 km/h	Medium
63	clinic	Krishiv's Dental Care & Orthodontic Centre	11.2184886, 78.1713346	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
17	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
18	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
30	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
43	school	Government Girls Higher Secondary School	11.2179186, 78.1672383	30 km/h	Medium
64	school	Namakkal Government Boys Higher Secondary School	11.2150472, 78.1669273	30 km/h	Medium
65	school	Maha Bharathi School	11.2099288, 78.2027271	30 km/h	Medium
66	school	Bharathi School	11.2108692, 78.2183269	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot

Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38295, 77.90001



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22407, 78.16640



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.21816, 78.16934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.20105, 78.23198



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.16124, 78.28095



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15975, 78.28885



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16047, 78.28943



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16056, 78.28971